

Regulator-Ready Forensic Checklist

Self-Assessment Framework for Deterministic AI Data Integrity

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Reference Standard	IAVP v1.0
Classification	Internal Assessment

Instructions

For each control requirement, assess your current implementation state:

Response	Definition	Risk Points
Yes	Control is fully implemented and documented	0
Partial	Control exists but implementation is incomplete or undocumented	1
No	Control is not implemented	2

Section I — Determinism & Replay

ID	Control Requirement	Yes	Partial	No	Risk
DR-01	The system SHALL produce identical outcomes when re-running a prior batch under identical configuration.	■	■	■	H
DR-02	Replay variance SHALL be explicitly measured and recorded per batch.	■	■	■	H
DR-03	The replay method SHALL be documented (e.g., full reprocess, partial, cached).	■	■	■	M
DR-04	Configuration SHALL be hashed and versioned for each batch.	■	■	■	H
DR-05	The number of replay runs SHALL be recorded (e.g., replay_runs = 3).	■	■	■	M

Section I Gap Count: _____ (Partial or No responses)

Section II — Cryptographic Attestation

ID	Control Requirement	Yes	Partial	No	Risk
CA-01	Batch results SHALL be signed using ECDSA P-256 or equivalent asymmetric algorithm.	■	■	■	H
CA-02	The signed artifact SHALL be independently verifiable without vendor access.	■	■	■	H
CA-03	The public key fingerprint SHALL be published and retrievable.	■	■	■	M
CA-04	Signature encoding SHALL be standardized (e.g., DER).	■	■	■	L
CA-05	The signing key SHALL be managed via HSM or cloud KMS.	■	■	■	H

Section II Gap Count: _____ (Partial or No responses)

Section III — Ordering & Integrity

ID	Control Requirement	Yes	Partial	No	Risk
OI-01	Input record ordering SHALL be deterministic and documented.	■	■	■	H
OI-02	Timestamps SHALL be canonicalized to a standard format (e.g., RFC 3339 UTC).	■	■	■	M
OI-03	UTF-8 normalization SHALL be enforced (e.g., Unicode NFC).	■	■	■	M
OI-04	Hash chaining SHALL be implemented across records.	■	■	■	H
OI-05	A root hash SHALL be produced per batch.	■	■	■	H
OI-06	The hash chain method SHALL be versioned (e.g., SHA256_CHAIN_V1).	■	■	■	M

Section III Gap Count: _____ (Partial or No responses)

Section IV — Failure & Enforcement

ID	Control Requirement	Yes	Partial	No	Risk
FE-01	The system SHALL fail closed on integrity errors.	■	■	■	H
FE-02	Demonstration keys SHALL be rejected in production environments.	■	■	■	H
FE-03	artifact_mode SHALL be enforced at runtime (PRODUCTION_REAL vs DEMO_SIMULATED).	■	■	■	H
FE-04	Partial or incomplete batches SHALL be rejected rather than silently accepted.	■	■	■	H
FE-05	Structural integrity violations SHALL be logged with timestamps.	■	■	■	M

Section IV Gap Count: _____ (Partial or No responses)

Section V — Retention & Governance

ID	Control Requirement	Yes	Partial	No	Risk
RG-01	Evidence SHALL be stored under WORM-compliant retention policy.	■	■	■	H
RG-02	Attestation artifacts SHALL be versioned and immutable once created.	■	■	■	H
RG-03	Prior protocol versions SHALL be preserved and accessible.	■	■	■	M
RG-04	Access to attestation materials SHALL be role-restricted.	■	■	■	M
RG-05	Retention period SHALL be documented (e.g., 7-year minimum).	■	■	■	M
RG-06	Legal hold capability SHALL be implemented.	■	■	■	H

Section V Gap Count: _____ (Partial or No responses)

Gap Analysis Summary

Section	Controls	Gap Count	High-Risk Gaps
I. Determinism & Replay	5		
II. Cryptographic Attestation	5		
III. Ordering & Integrity	6		
IV. Failure & Enforcement	5		
V. Retention & Governance	6		
TOTAL	27		

Risk Classification

Total Gaps (Partial + No)	Risk Level
0–5	Aligned
6–10	Partial Alignment
11+	Deficient

Assessment Result

Total Gap Count	
Risk Level	
Assessment Date	
Assessor	

Organizations with more than three "No" responses SHOULD evaluate the need for independent forensic review.

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